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A Machine Learning Perspective on the Brønsted–Evans–Polanyi Relation in Water-Gas Shift Catalysis on MXenes

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ABSTRACT

Machine learning (ML) has been employed to predict activation energies for key reactions in the water-gas shift process on pure MXene surfaces, namely, the dissociations of water (H₂O) and hydroxyl (OH), and the formation reactions of hydrogen (H₂) and carbon dioxide (CO₂). The dataset includes 92 MXenes, including single-transition-metal (M₂N or M₂C) and dual-transition-metal (M'₂M''C₂) compositions, with energies derived from density functional theory calculations from this work or the literature. Several ML models are evaluated, including Random Forest Regressor (RFR), Gradient Boosting Regressor, Artificial Neural Networks, Support Vector Machines, Decision Tree Regressor, and K-Nearest Neighbors Regressor, identifying RFR as the most accurate. Correlation plots comparing actual versus predicted activation energies further validate the robustness and accuracy of our models. Feature importance analysis reveals that two key factors, the reaction energy and the logarithmic partition coefficient (LogP) of the reactant, are crucial in predicting activation energies. While the former feature aligns well with the Brønsted–Evans–Polanyi principle relating to the kinetics and thermodynamics of a given reaction, the latter encompasses different physicochemical properties in only one parameter: polarity, hydrogen bonding, molecular size, molecular shape, and charge. This study underscores the power of ML in catalysis, offering a predictive framework to guide catalyst design.

1 | Introduction

The need for sustainable and efficient energy solutions has driven interest in materials capable of catalyzing key chemical reactions. MXenes, a class of two-dimensional (2D) transition metal carbides and nitrides, have emerged as promising candidates due to their distinctive physicochemical properties [1–6]. Since their discovery in 2011, they have drawn attention for their high electrical conductivity, hydrophilicity, and tunable surface chemistry, enabling applications in energy storage, sensing, and catalysis [7–11]. MXenes are synthesized by selectively etching A-layers from MAX phases, which are layered ternary carbides

or nitrides with the general formula M_{n+1}AX_n, where *n* is an integer (1–4), M is an early transition metal, A is typically an element from groups 13 or 14 of the Periodic Table, and X is carbon and/or nitrogen) [12–15]. The resulting 2D structure usually retains the metallic properties of the transition metal while exhibiting enhanced surface reactivity due to terminal groups such as -OH, -O, and -F [16–18]. This unique structure provides an extensive surface area and abundant active sites, which are crucial for catalytic reactions [19, 20]. Additionally, the tunable surface chemistry of MXenes allows for modification of their electronic properties, optimizing catalytic activity for specific reactions. This tunability is particularly important in reactions

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involving small molecules like H_2 , OH, H_2O , and CO_2 , which are central to energy conversion and storage technologies, due to their unique reactivity and interaction with catalysts.

Hydrogen production and storage, water splitting, and CO_2 reduction are pivotal processes in renewable energy systems [21–23]. Hydrogen, as a clean energy carrier, requires efficient catalysts for both its production (via water splitting) and storage [24, 25]. Similarly, converting CO_2 into useful chemicals and fuels addresses both energy supply and environmental concerns. MXenes, with their high conductivity and active surface site density, offer a promising platform for these catalytic processes [26–30]. Understanding the interaction between MXenes and adsorbates at the atomic level is essential for optimizing catalytic performance. Activation energy, which determines the rate of chemical reactions, is a critical parameter in this context. The ability to predict an activation energy accurately can significantly enhance the design and development of MXene-based catalysts [31].

Recent advances in machine learning (ML) have revolutionized various fields, including materials science [32–34]. ML techniques can uncover complex patterns and relationships within large datasets, enabling highly accurate predictions of material properties and performance [35, 36]. In catalysis, ML models can predict activation energies, reaction rates, and identify key descriptors that influence catalytic activity [37–40]. For instance, Roy et al. [41] used machine learning to predict the work function of MXenes. They developed models using basic chemical properties as features, trained on 275 data points from the Computational 2D Materials Database. Their neural network model achieved a mean absolute error (MAE) of 0.12 eV on training data and 0.25 eV on testing data. Feature importance analysis revealed that the electronegativity of surface-terminating atoms strongly influences the work function. Reduced-order models with fewer features demonstrated effective transferability to new materials, predicting the work function of MXenes with novel surface terminations like Br, Cl, S, N, and NH. This approach facilitates the rapid identification of tailored MXenes with targeted electronic properties for specific applications. Similarly, Abraham et al. [42] combined high-throughput quantum mechanical screening with artificial intelligence to identify key descriptors for CO_2 activation on MXenes. Their machine learning models screened 114 pure and defective MXenes, with the RFR model showing the best performance in predicting CO_2 adsorption energy. The RFR model demonstrated a mean absolute error of 0.16 eV for training data and 0.42 eV for testing data. Feature importance analysis identified the d-band center, surface metal electronegativity, and the number of valence electrons of the metal atoms as crucial descriptors. This approach highlights the potential of machine learning to guide the design of novel MXene-based catalysts by predicting key indicators for CO_2 activation.

Several studies have explored the adsorption and dissociation of molecules like H_2 , H_2O , and CO_2 on MXene surfaces. For example, Wang et al. [43] investigated hydrogen adsorption and dissociation on a two-dimensional Ti_2C monolayer. Although they found that hydrogen dissociates spontaneously on Ti_2C , it is important to note that MXenes without surface functionalization are challenging to stabilize. Additionally, Tahini et al. [44] studied water adsorption on V_2CO_2 , Ti_2CO_2 , and Nb_2CO_2 and despite

the respective adsorption energies were -0.23 , -0.15 and -0.49 eV, the addition of the zero-point energy and entropic contributions of the order of 0.67 eV resulted in positive Gibbs energy values. However, the incorporation of single metal atoms anchored on these MXenes, to yield, for example, $Fe@V_2CO_2$, $Mn@Ti_2CO_2$, and $Ir@Nb_2CO_2$, increased the stability of adsorbed water, which contributed to its relatively easy dissociation. Jurado et al. [45] examined nitride MXene surfaces without terminations for CO_2 capture using first-principles calculations. Ti-, Hf-, and Zr-based nitride MXenes showed the most exothermic adsorptions. Parey et al. [46] employed density functional theory (DFT) to investigate the potential of $Zr_2X(OH)_2$ ($X = C, N$, or B) MXenes for CO_2 conversion to formic acid via a surface hydroxylation mechanism. This involves a two-step coupled proton–electron transfer, where CO_2 is hydrogenated using hydrogen atoms from the OH-terminated MXene surface. Their results indicated that hydrogen abstraction from the surface is facile, effectively bypassing thermodynamic limitations.

To accelerate discovery beyond traditional DFT methods, several recent studies have employed machine learning to explore the catalytic potential of MXenes. For example, Yang et al. [47] combined ML with high-throughput DFT to investigate non-metal (namely, P, S, Se, Te) doped V_2C -based MXenes with different surface terminations (O, S). Their findings showed that dopant-induced modifications in the p_z orbital electron density significantly influence the Gibbs free energy of hydrogen adsorption (ΔG_{H^*}), enabling accurate hydrogen evolution reaction (HER) activity predictions across diverse compositions. They developed a universal ML-derived descriptor ($R^2 > 0.93$) that identified $P-V_2CTe_2$ as a top-performing catalyst with better ΔG_{H^*} than Pt and high thermal stability [47]. Moses et al. [48] introduced a robust ML/DFT multistep framework to assess the HER activity of over 4,500 $MM'XT_2$ -type MXenes. Using gradient boosting regression with recursive feature elimination, their model predicted ΔG_{H^*} with a low mean absolute error of 0.358 eV. Notably, Nb-, Mo-, and Cr-based MXenes with oxygen terminations emerged as highly active HER catalysts, surpassing commercial Pt-based systems. The work by Moses et al. [48] underscores the efficiency of ML in navigating the vast compositional design space of MXenes for hydrogen generation. In a separate ML enhanced screening, Xu et al. [49] examined 78 doped Janus multimetal $TiVCO_2$ MXenes, employing a classification model trained on DFT-derived ΔG_{H^*} values. Their 7-feature ML model achieved a prediction accuracy of 93.6% and identified five high-performing S-doped catalysts, subsequently validated by DFT. Importantly, this study of Xu et al. [49] illustrates the value of combining ML classification with DFT for efficient catalyst discovery, especially in systems with limited training data.

In this work, machine learning algorithms such as Random Forest Regressor (RFR), Gradient Boosting Regressor (GBR), Artificial Neural Networks (ANN), Support Vector Machines (SVM), Decision Tree Regressor (DTR), and K-Nearest Neighbors Regressor (KNN) were effectively employed to model and predict the activation energies of dissociation and association reactions. These models handle high-dimensional data, capture nonlinear relationships, and provide insights into the underlying mechanisms governing material behavior. The primary objective of this work is to apply machine learning techniques to predict

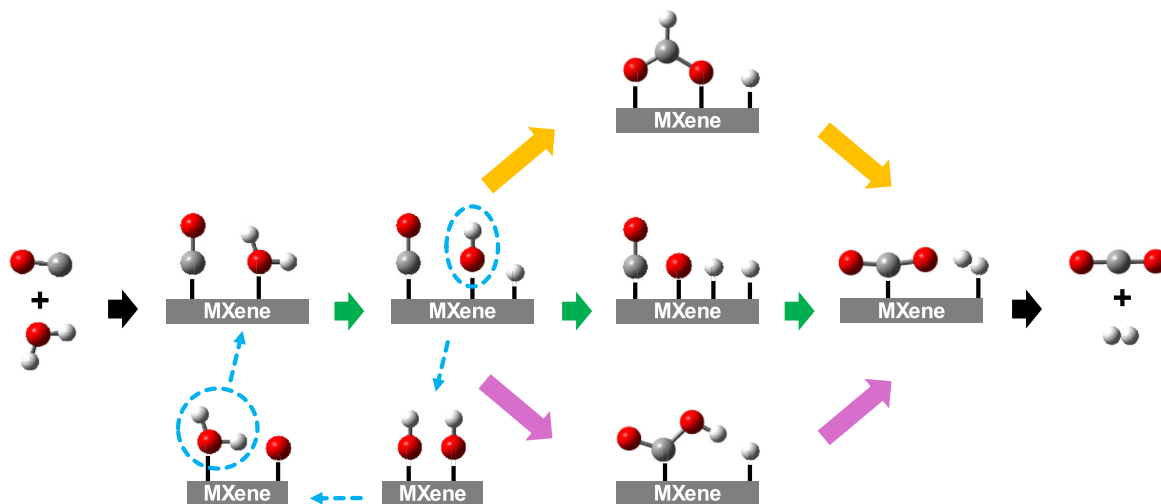


FIGURE 1 | Formate (top), redox (middle), and carboxylate (bottom) routes for the water-gas shift reaction. Green arrows show the elementary steps considered in this work while black arrows show the (reverse) adsorptions considered in our previous machine learning study [50]. The elementary steps in the formate and carboxylate routes as well as the disproportionation reaction of hydroxyl moieties, highlighted with yellow, magenta, and blue arrows, respectively, were not considered in this work. Color code for spheres: White is H; grey is C; and red is O.

the activation energies for H_2O and OH dissociation reactions as well as H_2 and CO_2 association reactions on MXene surfaces. These reactions are fundamental to the water-gas shift (WGS) reaction ($\text{H}_2\text{O} + \text{CO} \rightarrow \text{H}_2 + \text{CO}_2$, Figure 1), a critical process in hydrogen production [51–54]. Understanding the activation energies of these fundamental reactions is crucial for improving catalytic processes and designing more efficient catalysts for the WGS reaction.

Herewith, we utilized a dataset of 92 data points for various MXenes, derived from DFT calculations performed in this work and from the existing literature. This dataset includes critical properties such as lattice parameter, interlayer distances, Bader charges, and band gaps. Using Python packages like Matminer and RDKit, we calculated additional properties of MXenes and species relevant in the redox pathway of the WGS reaction, thereby enriching the dataset for a more robust analysis. We applied several ML to identify the most effective predictors of activation energy, focusing on evaluating their performance through cross-validation (CV) and a final test. Additionally, we analyzed feature importance of uncovering the key properties that influence the activation of energy.

2 | Methodology

2.1 | Dataset

The dataset used in this study consists of 92 data points representing various MXene properties, including their interactions with the molecules of interest: H_2O , OH , H_2 , and CO_2 . These properties were obtained from two primary sources: DFT calculations for 10 (bimetallic) MXenes, and literature data [17, 55, 56] for the remaining (single metal) MXenes. At this point it is important to add that the data, either computed in this work or taken from literature, were calculated with the identical computational parameters (e.g. same exchange-correlation functional, cutoff).

Therefore, potential concerns regarding data homogeneity are largely mitigated in this context.

The complete dataset, which includes all relevant properties, interaction data, as well as the codes and machine learning models used in this study, is available online at (<https://doi.org/10.17632/4zmkpw3wxx.2>). The dataset includes 28 data points for H_2O dissociation ($\text{H}_2\text{O} \rightarrow \text{HO} + \text{H}$), 18 data points for OH dissociation ($\text{OH} \rightarrow \text{O} + \text{H}$), 28 data points for H_2 association ($2 \text{H} \rightarrow \text{H}_2$), and 18 data points for CO_2 formation ($\text{CO} + \text{O} \rightarrow \text{CO}_2$).

As machine learning applications in MXene-based catalysis are still in their early stages, this study focuses on bare (non-functionalized) and defect-free MXene surfaces to isolate the intrinsic effects of core transition-metal chemistry. This approach enables a systematic and unbiased comparison across a wide range of MXene compositions by eliminating variability introduced by surface terminations and defects. While real-world MXenes are often terminated with functional groups such as $-\text{O}$, $-\text{OH}$, and $-\text{F}$, and may exhibit structural imperfections like atomic vacancies or substitutions [18, 57–60], excluding these factors provides a valuable baseline for uncovering fundamental activity trends. This controlled framework ensures data consistency, which is essential for training reliable machine learning models and identifying general structure–property relationships. Beyond the predictive power of our results, this approach lays the groundwork for future studies that incorporate surface terminations and defects to explore how these features modulate activation energies and influence catalytic performance under experimentally relevant conditions.

2.2 | DFT Computational Details

For 10 MXenes with the formula $\text{M}'_2\text{M}''\text{C}_2$, where M' is Cr, Mo, or Ti, and M'' is Nb, Ta, Ti, or V, DFT calculations were conducted to obtain data such as activation energies, Bader charges, lattice parameters, and other critical descriptors related to

MXenes. These parameters, which are essential for understanding the catalytic properties of MXenes, are listed in Table S1 in the Supporting Information. The calculations were performed using the Vienna Ab Initio Simulation Package (VASP) [61]. Specifically, the Perdew–Burke–Ernzerhof (PBE) exchange–correlation functional was employed, a widely used Generalized Gradient Approximation (GGA) functional that effectively represents the electronic structure and energetics of various materials, including MXenes [62]. To account for long-range van der Waals interactions, D3 dispersion corrections were included in the calculations [63]. This approach provides a robust treatment of adsorption phenomena suitable for both metallic and non-metallic systems. The MXenes studied in this work, with the formulas M_2C , M_2N , and $M'_2M''C_2$, were modeled with 16 atoms per layer to represent their surfaces, that is, using p (4×4) periodic supercell structures, as illustrated in Figure 2, and $3 \times 3 \times 1$ k -point grids within the Brillouin zone [64]. Isolated molecules were modeled in asymmetric boxes with dimensions $10 \times 11 \times 12$ Å³, using a single k -point (the Γ -point). Convergence criteria for the self-consistent field (SCF) and ionic relaxation steps were set at 10^{-6} eV and 0.005 eV/Å, respectively, to ensure sufficient accuracy. The plane-wave energy cutoff was set to 550 eV, a standard value for these calculations. Projector augmented wave (PAW) pseudo-potentials were used for all elements in the system, and ionic relaxation was performed using the conjugate gradient algorithm [65].

The activation energy (E_a) quantifies the energy required for a reaction, either it is for the dissociation of an adsorbed molecule into its respective fragments or the association of fragments to form a molecule. In this study, we focus on the dissociation reactions of water,

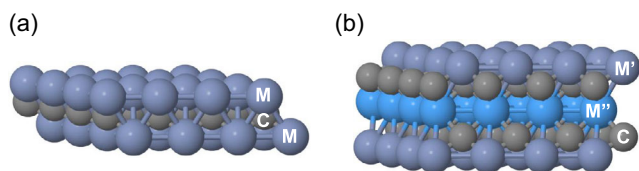


FIGURE 2 | Supercell structures of MXenes: (a) M_2C and (b) $M'_2M''C_2$. The M_2C structure is representative of both M_2C and M_2N , as their configurations are structurally similar. The layers in each structure are labeled as M (metal) and C/N (carbon or nitrogen).



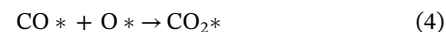
where $*$ denotes an adsorption site on the MXene surface, and hydroxyl,



as well as the association reactions of hydrogen adatoms



and carbon monoxide with oxygen



catalyzed by MXenes. These reactions are directly relevant for the redox route of the WGS reaction [54], Figure 1. In the future, we plan to obtain data for key intermediate species involved in the carboxylate and formate pathways, as well as potential competing reactions, to develop a more comprehensive model.

For the dissociation reactions, the activation energy (E_a) is calculated as the difference between the total energy of the transition state, $E_{\text{transition}}$, and the total energy of the initial state (i.e., H_2O or OH adsorbed on the MXene surface):

$$E_{a(\text{dissociation})} = E_{\text{transition}} - E_{\text{initial}} \quad (5)$$

For association reactions, the activation energy is related to the activation and reaction energy of the corresponding dissociation reactions (i.e., reverse reactions) as:

$$E_{a(\text{association})} = E_{a(\text{dissociation})} - E_{r(\text{dissociation})} \quad (6)$$

where $E_{a(\text{dissociation})}$ is the activation energy for the corresponding dissociation reaction, and $E_{r(\text{dissociation})}$ is the dissociation reaction energy. Low activation energy indicates that the MXene surface catalyzes the reactions, either bond breaking (dissociation) or bond formation (association), more effectively, hence the corresponding MXene surfaces are better catalysts.

Figure 3 highlights critical catalytic reaction pathway configurations, namely, (a) initial state $\rightarrow$ (b) transition state $\rightarrow$ (c) final state, using the water dissociation on an $M'_2M''C_2$ MXene surface

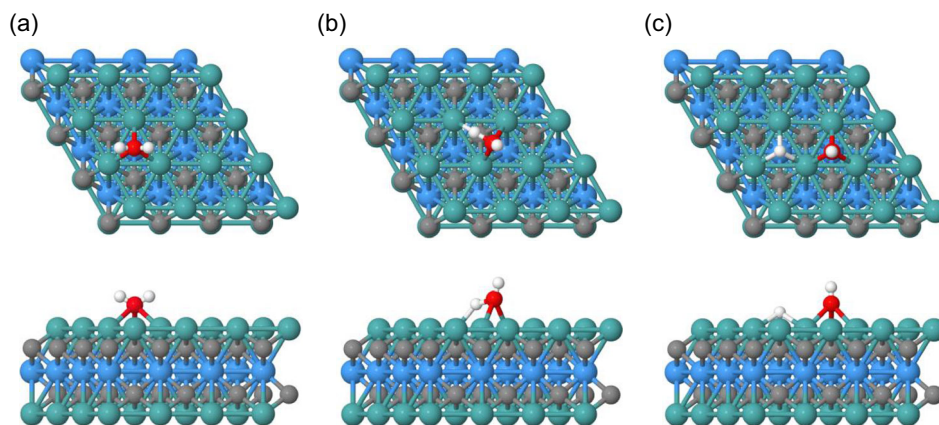


FIGURE 3 | Top and side views of the (a) initial, (b) transition, and (c) final states of the water dissociation reaction on a $M'_2M''C_2$ MXene surface.

as an example. The structure in panel (a) shows the optimized structure of a water molecule interacting with the surface through physisorption or weak chemisorption. The adsorption energy of water is calculated from the total energy of the MXene with adsorbed water and the energies of the separated fragments, that is, the gas-phase H_2O molecule and the MXene model surface. Panel (b) depicts the transition state, which represents the highest-energy point along the reaction coordinate, where an O–H bond is stretched. The energy difference between the structures in panels (b) and (a) corresponds to the activation energy of the water dissociation reaction, with more active surfaces having lower activation energies than less active surfaces. Panel (c) presents the final state, with dissociated H and OH fragments adsorbed at distinct sites. The energy difference between the structures in panels (c) and (a) corresponds to the energy of the reaction (negative/positive values are exothermic/endothermic reactions).

2.3 | Feature Engineering

To enhance the dataset and capture the complex interactions between MXenes and adsorbates, we obtained additional features utilizing three Python libraries: Matminer, Mendeleev and RDKit. Matminer was used to derive material features [66], whereas Mendeleev was used to obtain elemental properties related to the MXene atoms. On the other hand, RDKit [67], a cheminformatics library, was used to calculate molecular descriptors for the main adsorbate species of each reaction, including H_2O , OH, H_2 , and CO_2 . These features span a variety of categories, each obtained using different methods. The chemical composition and molecular structure of the materials were analyzed to derive features such as electron affinity, ionization properties, band center, and thermal conductivity using Matminer. For the main adsorbate species of each reaction (H_2O , OH, H_2 , and CO_2), properties like molecular weight, partial charges, radical electrons, and valence electrons were obtained using RDKit. The characteristics of the transition metals in MXenes were determined using Mendeleev, including elemental properties such as atomic radius, atomic weight, melting point, electronegativity, and first ionization energy. Additionally, structural, electronic, and thermodynamic properties of MXenes were derived from DFT calculations, including bond lengths, layer distances, Bader charges, and formation energy. Surface and reactivity properties were also investigated

using DFT, such as the adsorption energy of the reactants (in the reactions described in Equations (3) and (4), the adsorption energies were evaluated as the sum of CO and O, or two H atoms, respectively), reaction energy and activation energy. This combination of methods ensures a comprehensive set of features that captures the essential properties relevant to the prediction of activation energies. In total, we generated 103 features, which are summarized together with their descriptions in Table S2 in the Supporting Information, to feed into our models. Integrating these calculated features into the models allows them to recognize complex patterns and relationships within the dataset, identifying which features are most important and influential in predicting the activation energy.

We employed recursive feature elimination (RFE) for feature selection, to identify the most relevant features for prediction, while eliminating redundant or less informative ones. By systematically removing the least important features, RFE refines the model to improve its performance and interpretability. Features with lower importance are discarded, and the model is retrained using the remaining ones. This iterative process continues until only the essential features are retained. In this work, RFE was combined with cross-validation and mean absolute error to rank and select the most important features. The final set of features, after multiple iterations, represents the most important ones for making accurate predictions, thus improving both the accuracy and interpretability of our model.

2.4 | Machine Learning Models

In Figure 4, we illustrate the workflow of the machine learning process employed in this study to predict the activation energies for the dissociation of H_2O , OH, and for the association reactions of H_2 and CO_2 on MXene surfaces. The process begins with an analysis of MXene structures, followed by data collection from DFT calculations and literature reference works. The collected data is then processed through feature engineering to identify key descriptors, with correlations visualized using heatmaps. ML models are subsequently trained and evaluated to select the best-performing model and optimized hyperparameters according to cross-validation, which is then tested for accuracy and generalization using a final test set with holdout data used just for this purpose. This workflow underscores the integration of computational and machine learning techniques in MXenes

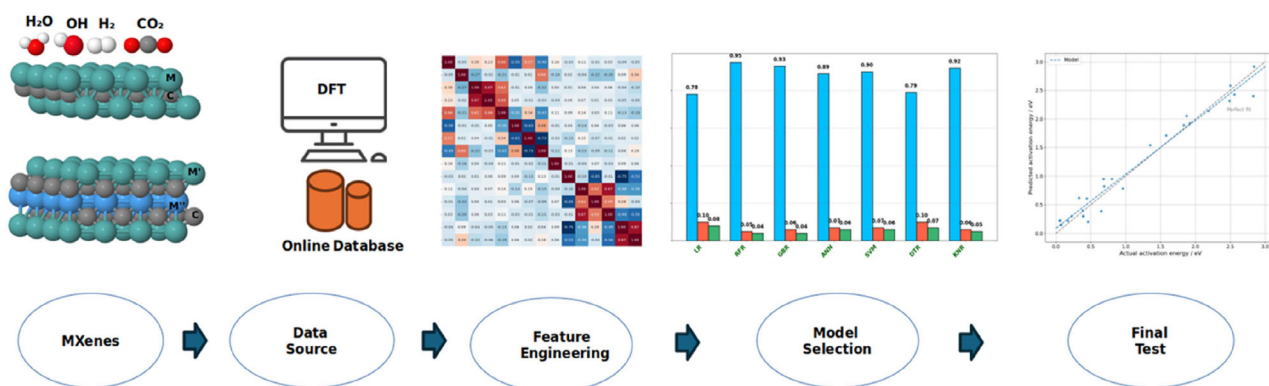


FIGURE 4 | Workflow of machine learning in this study.

research, with the model development described in detail in the next few subsections.

In this study, several ML models were employed, each with its distinct advantages. The RFR was used due to its capability to handle large, complex datasets and effectively capture nonlinear relationships [68]. ANNs were selected for their ability to model intricate patterns through a multilayer structure. ANNs are well-suited for high-dimensional data, as they can detect complex relationships between variables by mimicking the way the human brain processes information [69]. To further enhance prediction performance, GBR was considered for its efficiency in employing boosting techniques. It works by combining weaker models into a stronger one and is renowned for its high accuracy and ability to handle noisy data [70]. SVMs were also applied due to their effectiveness in finding optimal hyperplanes for regression tasks. SVMs can handle both linear and nonlinear data, making them versatile for different types of problems [71]. DTR was included for its intuitive nature, providing clear interpretations by splitting the dataset into branches based on feature values. Though they can be prone to overfitting, decision trees are easy to visualize and understand [72]. Additionally, the k-nearest neighbors regressor was chosen for its simplicity and effectiveness, particularly for smaller datasets. This model predicts the output based on the nearest data points, making it a straightforward but powerful tool for regression tasks [73]. In addition to these models, multivariable linear regression (MLR) was also applied due to its simplicity and interpretability. MLR assumes a linear relationship between the predictors and the response variable, making it a valuable baseline model for comparing with more complex machine learning models [74].

By implementing these models using Python's scikit-learn library [75, 76], the study aimed to compare and identify the most effective models, ultimately ensuring robust and reliable predictions of activation energies in the context of MXene-catalyzed reactions. To guarantee a thorough evaluation and accurate prediction of activation energies, we implemented a structured approach involving training, validation, and testing. The models were first trained and validated by means of cross-validation using 70% of the dataset (64 data points), which played a key role in optimizing the model and mitigating overfitting. The remaining 30% (28 data points) were kept aside as a test set to evaluate the performance of the model on previously unseen data.

2.5 | Model Tuning and Evaluation

In this work, we followed a rigorous methodology for model training and evaluation to ensure the accuracy and reliability of activation energy predictions. Each model was thoroughly assessed using 5-fold cross-validation, where the dataset was divided into five subsets for iterative training and validation. We employed several models to better understand the relationship between input features and activation energies. To fine-tune our models, we applied hyperparameter optimization using RandomizedSearchCV, with all details provided in the Supporting Information. This technique involves randomly sampling hyperparameter combinations and evaluating their performance. For evaluating the regression models, we utilized key metrics including R-squared or coefficient of determination (R^2), mean absolute error (MAE), and root-mean-squared error

(RMSE). This comprehensive approach enabled us to refine our models for enhanced predictive accuracy and better generalization to new data.

2.6 | Final Test

At this stage, the model was evaluated using a test set comprising 30% of the total dataset. This subset was withheld during the training and validation phases to ensure an unbiased evaluation. The performance of the final model was assessed on this unseen data, providing insights into its ability to generalize and accurately predict activation energies for reactions involving MXene materials. These results were critical for verifying the predictive accuracy and reliability of the model.

3 | Results and Discussion

3.1 | Distribution and Analysis of Activation Energies

Before predicting and analyzing activation energies, it is crucial to understand their distribution and variation across the dataset, as this provides key insights about the potential of MXenes for application in catalysis. Figure 5 offers a detailed analysis of the density and frequency distributions of activation energies for the dissociation reactions of H_2O and OH , as well as the association reactions leading to H_2 and CO_2 formation on MXene surfaces. Figure 5a presents the density distributions of activation energies for the H_2O and OH dissociation reactions and for the association reactions originating H_2 and CO_2 . The activation energies for H_2O (blue) are primarily concentrated at low values, peaking around 0.44 eV, suggesting that H_2O dissociation reactions generally require less energy compared to other adsorbates. For OH (green), the distribution is sharply peaked around 0.74 eV, indicating a narrow range of activation energies, meaning that OH related dissociation reactions exhibit relatively uniform energy requirements. In contrast, the activation energy distribution for H_2 (red) is much broader, spanning over 3 eV, reflecting significant variability in the energy required for H_2 association, likely due to differing surface interactions or multiple reaction pathways. Similarly, for CO_2 (cyan), the activation energies are also rather widely spread, peaking around 2.3 eV with a long tail extending beyond 3 eV, indicating that CO_2 dissociation on the MXene surface is relatively energy-intensive and exhibits considerable variability. The overlap between these distributions suggests that some activation energies for different adsorbates fall within similar ranges, which may imply competition for adsorption sites or shared reaction pathways.

Figure 5b further supports these observations by displaying the frequency distribution of activation energies, offering a discrete count of occurrences. The activation energy values for H_2O are densely packed below 0.5 eV while those for OH are clustered between 0.5 and 1 eV. In the case of the reaction towards H_2 , the activation energies show a broader distribution, with frequent occurrences around 0.5–1 and 2–3 eV, suggesting multiple reaction pathways or variations in MXene surface interactions.

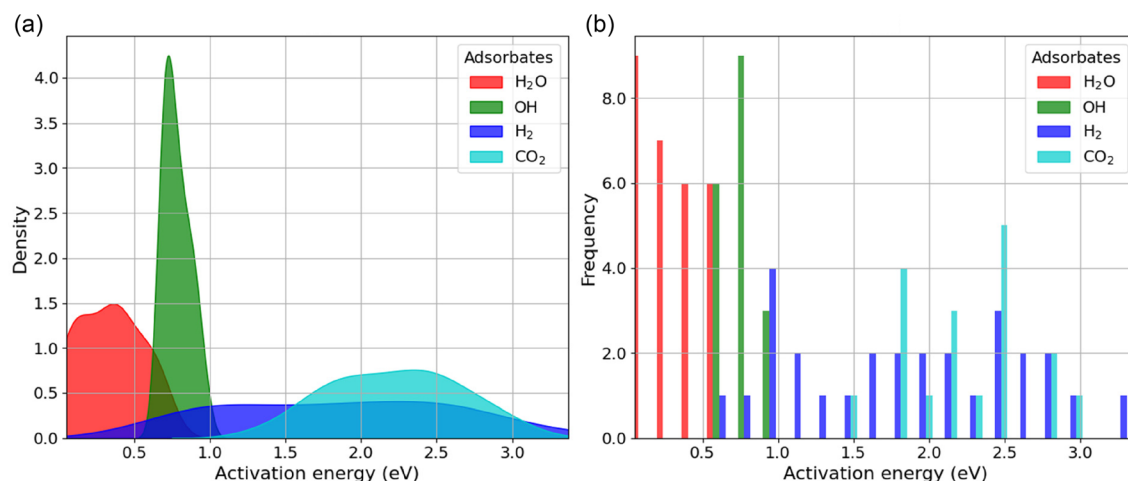


FIGURE 5 | Density and frequency distributions of activation energies. (a) Kernel density estimate plot and (b) Histogram. Labels H₂O, OH, H₂ and CO₂ stand for reactions described by Equations (1)–(4), respectively, that is, H₂O and OH dissociations, and H₂ and CO₂ formations.

Similarly, for the CO oxidation reaction to CO₂, the activation energies span the range 1.5–3 eV.

Previously, it was found that MXenes with d² configurations generally exhibit lower energy barriers towards H₂O dissociation, followed by d³ and then by d⁴, with tungsten-based MXenes being the least effective of the metals analyzed [56]. These insights emphasize the importance of compositional refinement in designing MXenes tailored for specific catalytic reactions. While an activation energy of ~0 eV would theoretically represent an ideal scenario for a spontaneous reaction, such cases are exceedingly rare in real systems [77]. In fact, activation barriers within a specific range are preferred for applications in practice, as they balance reactivity and selectivity, ensuring controlled and efficient catalysis [20].

3.2 | Exploring Feature Selection and Analysis

Feature selection plays a crucial role in enhancing model performance and interpretability by identifying the most impactful variables while reducing complexity and mitigating overfitting (Figure 6). Using Recursive Feature Elimination with Random Forests, two key features were identified from an initial pool of over 100 as the optimal set for predicting the activation energies of the reactions present in the dataset (dissociation reactions of H₂O and OH, and association reactions leading to H₂ and CO₂). The first selected feature is reaction energy, which represents the energy change occurring during a reaction, from reactants to products. This feature is crucial as it directly correlates with the thermodynamic feasibility and stability of reaction intermediates, influencing activation energy requirements. The second feature is LogP of the reactant, the logarithm of the octanol/water partition coefficient, which indicates how a reagent distributes itself between hydrophobic (lipid-soluble) and hydrophilic (water-soluble) phases. While this property may be surprising, it contributes with a measure of the hydrophilic nature of the adsorbing molecule, hence affecting indirectly adsorption behavior.

The hydrophilic nature of a molecule can depend on the interplay between its polarity, hydrogen bonding, molecular size,

molecular shape, and charge. Therefore, the LogP is a single descriptor, which has the ability to take into account different physicochemical properties within the same parameter, being selected using RFE among other descriptors which could have been used to identify the main molecule in each reaction in this work, such as molecular weight, topological polar surface area (TPSA), fingerprint molecular similarity calculated according to so-called Morgan density, maximum absolute partial charge, number of radical electrons, and number of valence electrons. The RFE process, guided by mean absolute error as the evaluation metric, incorporated cross-validation to ensure the robustness of feature selection. By systematically eliminating irrelevant or redundant features, this approach improved the predictive accuracy and generalizability of the model.

Additional features were incorporated in Figure 6 to investigate a broader set of relationships within the dataset, which would provide a deeper understanding of the overall system. These additional features include the molecular weight of the main adsorbate species (H₂O, OH, H₂ and CO₂) which affects their interactions with the MXenes; the maximum absolute partial charge of the main adsorbate species, which is related to their reactivity; and MXene-specific characteristics, such as the formation energy and adsorption energy of the reactants, which describe the stability and interaction strength between the MXenes and the reactants.

The distance between atoms within the MXene structure was considered as well, as it can be related to the electronic distribution of the MXene surface. Including these extra features, alongside the primary ones, allows for a more subtle exploration of the complex interactions between the properties of MXenes and the adsorbates. This approach provides a more comprehensive understanding of the factors influencing activation energy predictions, ensuring that the model captures not only the most critical predictors but also a broader view of the structure of the dataset and the intricate interplay between the properties of the materials and their catalytic behaviors. To better understand the relationships between the features, including the selected key ones, we analyzed the correlation matrix, which is visualized as a heatmap in Figure 6. The correlation coefficients range from +1

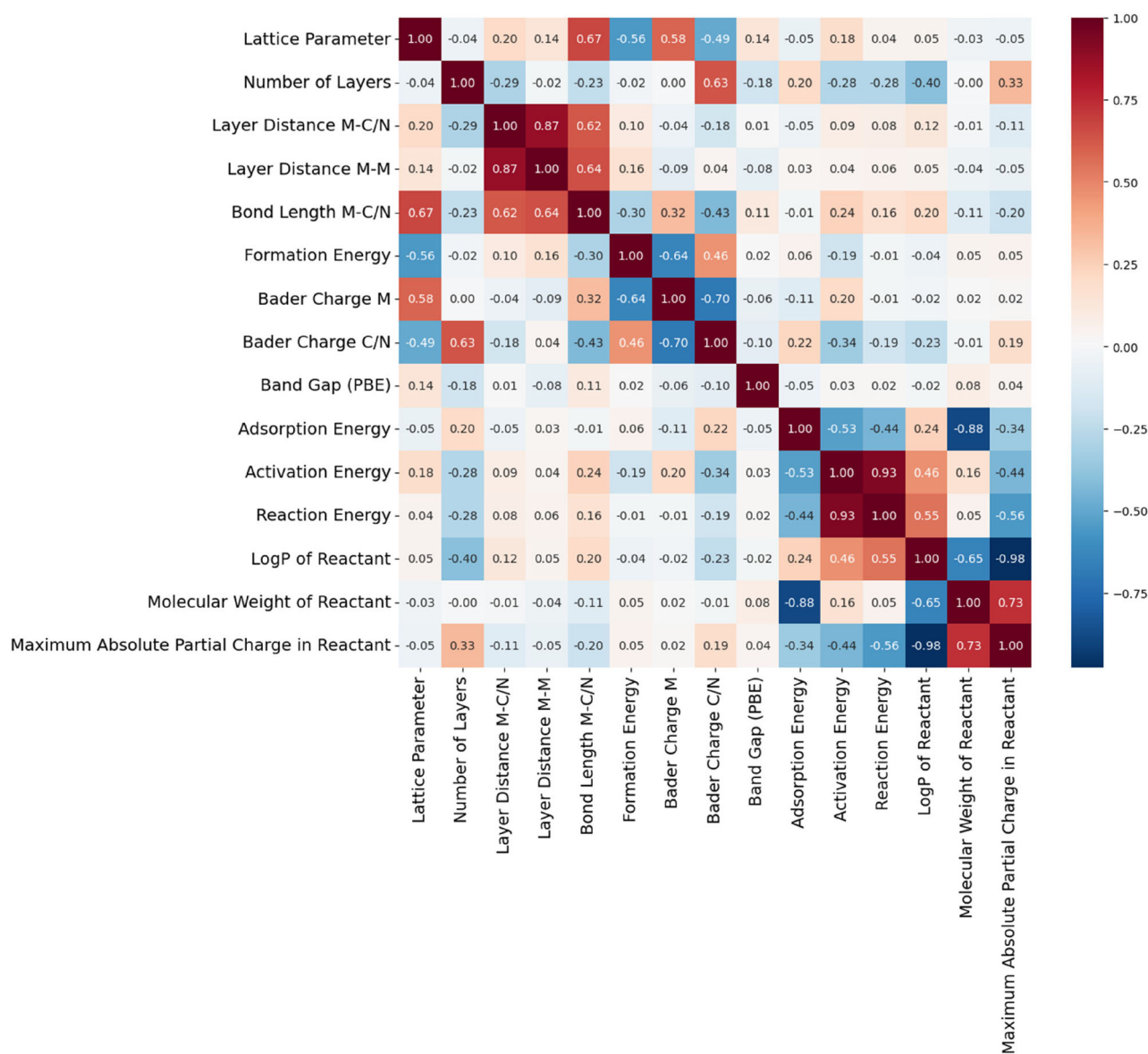


FIGURE 6 | Correlation matrix heatmap showing the relationships between selected features and other features studied in this work, as well as the properties of MXenes.

to -1 , where $+1$ (indicated by darker red) represents a perfect positive correlation, meaning that as one feature increases, the other also increases proportionally. Conversely, -1 (in darker blue) represents a perfect negative correlation, where an increase in one feature corresponds to a decrease in the other. Values near 0 (shown in white) suggest no significant linear relationship between the variables. This heatmap revealed important patterns that helped identify interdependencies and redundancies among features. Features with strong correlations may provide overlapping information, while those with weak correlations contribute more independently. This analysis supports the relevance of the selected features, ensuring they not only add value independently but also align with the underlying physical and chemical principles of the system.

As observed in Figure 6, the reaction energy shows a strong positive correlation with activation energy, that is, more exothermic/endothermic reactions tend to have lower/higher activation energies. This observation supports the Brønsted–Evans–Polanyi

principle in heterogeneous catalysis, demonstrating that reaction energy and activation energy are closely linked in catalytic processes [78, 79]. Adsorption energy of the reactants has a strong negative correlation with activation energy, suggesting that stronger adsorption of reactants onto the catalyst surface tends to result in higher energy barriers [80].

The strong negative correlation between LogP and maximum partial charge in reactants suggests that more hydrophobic molecules tend to have lower partial charges, likely due to the absence of strongly polar functional groups. However, given the small size of the molecules studied, this trend may be constrained by the limited variability in their structures. Similarly, while molecular weight shows a negative correlation with LogP, this does not necessarily imply a straightforward relationship between size and hydrophobicity, as molecular polarity is influenced by both local functional groups and overall charge distribution.

Please note that analyzing correlations, such as the one observed between molecular weight and adsorption energy, can be misleading if the broader context is overlooked or if only a limited set of species is considered. Adsorption mechanisms are highly complex and, therefore, such analyses must be conducted with care.

3.3 | Predictive Modeling of Activation Energy

We began our study by employing linear regression to predict activation energies. The initial results from cross-validation yielded an R^2 value of 0.79, indicating a moderate level of predictive capability. Nevertheless, the performance of the linear regression model can be limited by its inability to capture complex, non-linear relationships between the features and activation energies. Therefore, other machine learning methods, such as random forest regressor, SVM, and neural networks, were also tested because they are better equipped to handle the complexities of the data and are particularly effective at capturing the non-linear interactions that linear models often miss.

The cross-validation results for each model studied in this work are summarized in Figure 7. The non-linear models, except for DTR, exhibit a higher level of predictive accuracy. Furthermore, we evaluated the different models using varied random states for the data split between data used for cross-validation and the final test set, and the results remained consistent.

We developed an RFR model and evaluated its performance using 5-fold cross-validation. The performance metrics, including R^2 , RMSE, and MAE, along with their standard deviations, are summarized in Table 1. The low standard deviations further confirm the stability of the performance of the model. Overall, the RFR model demonstrates strong predictive capabilities, effectively modeling activation energies and generalizing well to new data, making it a reliable choice.

TABLE 1 | Cross-validation results for different models.

Model	Mean R^2	Mean RMSE (eV)	Mean MAE (eV)
LR	0.79 ± 0.13	0.34 ± 0.02	0.28 ± 0.03
RFR	0.95 ± 0.04	0.17 ± 0.06	0.12 ± 0.04
GBR	0.94 ± 0.04	0.19 ± 0.06	0.13 ± 0.03
ANN	0.90 ± 0.06	0.23 ± 0.02	0.19 ± 0.02
SVM	0.90 ± 0.06	0.23 ± 0.02	0.19 ± 0.02
DTR	0.79 ± 0.17	0.33 ± 0.16	0.25 ± 0.10
KNR	0.93 ± 0.03	0.21 ± 0.06	0.15 ± 0.04

To optimize the model, we employed scikit-learn's RandomizedSearchCV [81], a technique that identifies the best hyperparameters by randomly sampling from predefined values, rather than testing all possible combinations. This approach helps to efficiently explore the hyperparameter space while reducing computational cost. A more detailed description of how this method was applied in our work, along with the best parameters for the models, is provided in the Supporting Information. The optimization process led to improved performance, enhancing the accuracy and stability of the model in predicting activation energies. We evaluated model performance using key metrics: R^2 ; MAE, which calculates the average magnitude of errors between predicted and actual values; and RMSE, which gives greater weight to larger errors by evaluating the square root of the average squared differences between predicted and actual values [82].

The optimized hyperparameters were: {'n_estimators': 100, 'min_samples_split': 2, 'min_samples_leaf': 2, 'max_features': 'sqrt', 'max_depth': 30, 'bootstrap': False}. Under these conditions, the model achieved a mean cross-validation R^2 of 0.95 ± 0.04 , mean

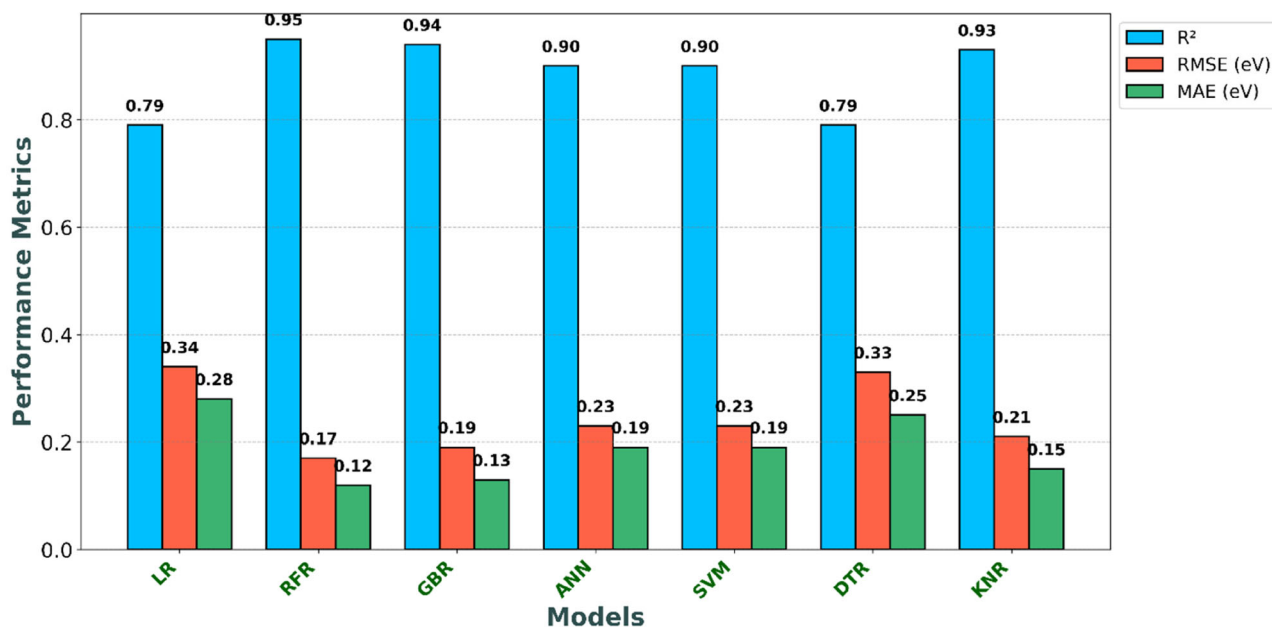


FIGURE 7 | Variation in performance results (R^2 , RMSE, and MAE) from cross-validation for RFR, GBR, ANN, SVM, DTR, and KNR models, compared to the linear regression model.

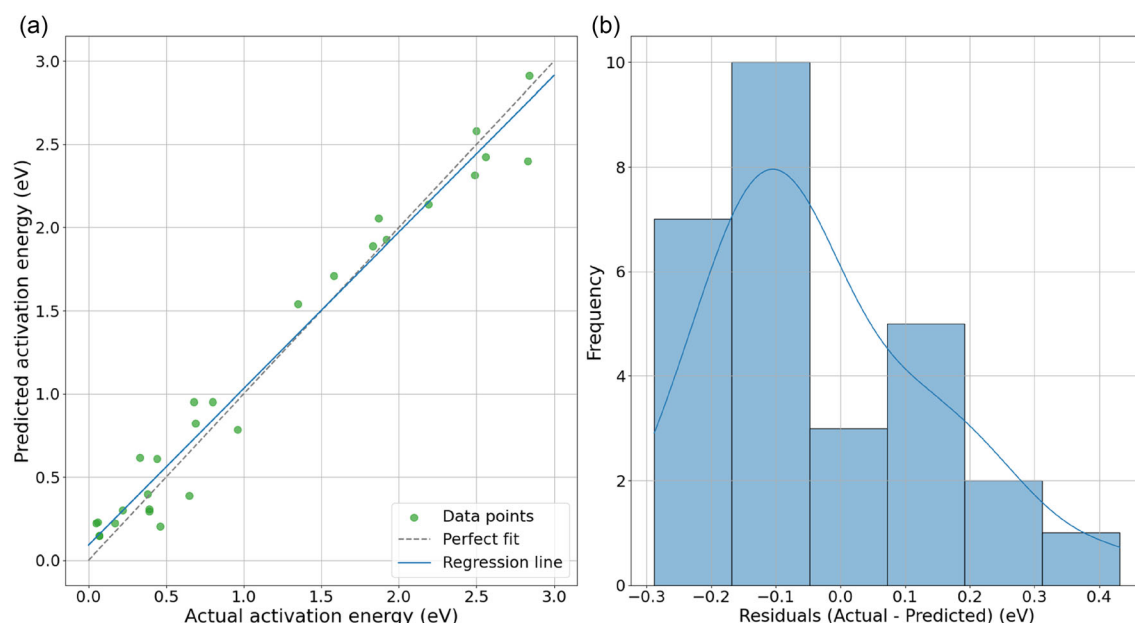


FIGURE 8 | Final test RFR model accuracy and residual analysis for predicted activation energies, (a) scatter plot of actual versus predicted energies, (b) residual Histogram.

RMSE of 0.17 ± 0.06 eV, and mean MAE of 0.12 ± 0.04 eV, respectively. These results demonstrate that the model performs well, exhibiting a high R^2 and low RMSE and MAE, highlighting the RFR model accuracy and its strong ability to predict activation energies effectively. After optimization and CV, we tested the RFR model on an independent dataset, yielding excellent results.

Figure 8 consists of two subplots for the final test: (a) a scatter plot comparing predicted vs. actual activation energy and (b) a histogram of residuals (Actual - Predicted values). In subplot (a), the data points closely align with the diagonal “Perfect Fit” line, indicating that the predictions of the model are highly accurate. The proximity of the points to the regression line further highlights a strong correlation between actual and predicted values, supported by a high coefficient of determination $R^2 = 0.97$, RMSE = 0.17 eV and MAE = 0.14 eV, demonstrating excellent predictive capability. This suggests that the model effectively captures the key descriptors influencing activation energy and can reliably be used for prediction on MXene surfaces. Subplot (b) presents the residuals, visualized as a histogram, showing the differences between actual and predicted activation energies. Most residuals are centered around 0 eV, confirming the strong predictive performance of the model, with most values falling between -0.3 and 0.3 eV. This tight distribution indicates that prediction errors are minimal and evenly distributed. The low magnitude of these residuals reinforces the reliability of the model, making it a robust tool for activation energy prediction in catalysis applications.

While the residuals in Figure 8b are largely concentrated within ± 0.3 eV, with only one residual falling outside this range, namely the Ta₂C surface for the CO + O $\rightarrow$ CO₂ reaction, with a residual of 0.43. This is a small percentage of the test set ($\sim 3.5\%$), with all the other examples following a normal error distribution. This dispersion suggests that the deviations may arise from complex or less-represented local environments within certain

MXene compositions. Although such instances exist, the very limited number and distribution do not indicate a systematic bias in the model. Importantly, their presence reinforces the value of expanding the dataset and incorporating more nuanced descriptors in future work, particularly to capture subtler electronic or structural variations that influence activation energy predictions.

We also explored the use of the GBR model, which demonstrated good performance in predicting activation energies, according to the following cross-validation performance metrics: $R^2 = 0.94 \pm 0.04$, RMSE = 0.19 ± 0.06 eV, and MAE = 0.13 ± 0.03 eV. However, when comparing the performance of the GBR model with the RFR model, we found that the RFR model slightly outperformed GBR in terms of R^2 . As a result, the RFR model was chosen for further analysis, particularly to explore the feature importance in the remainder of this study.

In addition, we evaluated other algorithms, including ANN, SVM, DTR, and KNR, to compare their performance. Based on the CV metrics summarized in Table 1, RFR consistently outperformed all other models. In contrast, besides LR, DTR showed the lowest mean R^2 , 0.79 ± 0.17 , and the highest variability in performance. It also recorded the highest RMSE of 0.33 ± 0.16 eV and MAE of 0.25 ± 0.10 eV, making it the least reliable and accurate model. Overall, RFR emerged as the most reliable and accurate model for predicting activation energies. It effectively captured complex nonlinear interactions and generalized well across the dataset, as evidenced by both 5-fold cross-validation and the test set. A recent study, which employed machine learning to predict the adsorption energies of various molecules (H₂O, CO₂, H₂, CO, O₂, OH, O, and H) on MXenes, also identified RFR as the most effective algorithm, further confirming its potential in this domain [50]. Notably, RFR outperforms alternative approaches such as gradient boosting, neural networks, and SVM. Its ensemble architecture and inherent robustness to overfitting make it particularly well-suited for

heterogeneous datasets like the ones used in this study. While RFR offers strong predictive performance, its limited interpretability at the level of individual predictions can hinder the extraction of detailed mechanistic insights. To overcome this, SHapley Additive exPlanations (SHAP) [83, 84] analysis was performed for understanding how specific features influence individual predictions (next section).

This study focuses on using ML to predict activation energies for various reactions involving water, hydroxyl, hydrogen, and carbon dioxide on MXene surfaces. MXenes, known for their versatile catalytic properties, are tailored to suit a range of chemical reactions, particularly those important in sustainable energy technologies. The study explores the relationship between surface catalytic properties, such as adsorption energies and charge transfer capabilities, and activation energies. Despite the limited dataset, the machine learning models showed promising predictive accuracy, with plans to expand the dataset and incorporate additional models for better performance. Future work will focus on increasing the dataset through computational simulations, including more reactions and surface species, and enhancing the interpretability and reliability of the models through methods like DFT simulations. These efforts aim to improve the predictive framework and support the development of scalable, generalized catalytic solutions.

In the study of Hutton et al. [85] which focused on predicting activation energies for chemical reactions on metal surfaces, the authors employed RFR, SVR, and GBR to predict activation energies for reactions involving C, O, and H-containing molecules on transition metal surfaces. They reported the following performance metrics: RFR with a mean absolute error of 0.17 eV, GBR with an MAE of 0.15 eV, and SVR with an MAE of 0.17 eV. In comparison, our work takes a similar approach by using machine learning models to predict activation energies, but with a focus on MXenes. Our results, however, surpassed the performance metrics presented by Hutton et al., with an MAE of 0.12 eV. This supports that our model is well suited for predicting activation energies on MXene surfaces, likely due to the unique properties of MXenes and the customized approach we adopted to account for their surface characteristics. Both studies highlight the superiority of non-linear models, such as RFR, over traditional linear regression, which struggles to capture the complexity of interactions in catalytic systems. The parallels between our work and that of Hutton et al., emphasize the growing role of ML in accelerating the understanding of complex catalytic systems.

The findings from our work demonstrate that ML models, particularly ensemble methods like RFR and GBR, can effectively predict activation energies even when applied to a relatively small dataset (92 data points). While smaller datasets are often considered a limitation in machine learning, we recognize that with careful model selection and data handling, effective predictions can still be achieved. To mitigate overfitting and ensure the robustness of our models, we employed several strategies, including 5-fold cross-validation, regularization techniques, and hyperparameter optimization through RandomizedSearchCV. Studies reported in the past year support the validity of machine learning approaches, even when applied to small datasets in catalysis research. For example, Taniike et al. [86] developed an automatic feature engineering method that extracts relevant descriptors without prior

domain-specific knowledge and successfully applied it to three different catalytic systems with limited data. Their results show that accurate ML models can still be developed under such constraints. Abraham et al. [87] reviewed the application of ML in catalysis and emphasized that even small, domain-specific datasets can yield valuable insights when coupled with appropriate algorithm selection and consistent data handling principles. Similarly, Chen et al. [88] introduced a small-data-oriented workflow for catalytic propane activation and found that ensemble models such as Random Forest and CatBoost are particularly effective in small data scenarios. These three examples reinforce that with careful model selection and data curation, as implemented in our study, both reliability and interpretability can be effectively maintained.

3.4 | Feature Importance and SHAP Analyses

Feature importance analysis is a crucial technique used to understand how individual features or variables influence the prediction outcome in a model. It helps pinpoint which factors have the greatest impact on the accuracy of the model, enabling a better understanding of the underlying data and its relationships [89, 90]. In our study, we employed feature permutation importance, a method that assesses the contribution of each feature by randomly shuffling its values and measuring the resulting change in the model's performance. A notable drop in performance indicates that the feature is highly important, while a minimal change suggests the feature has lower relevance to the predictions of the model. This technique can be applied to any model type and offers insights into the most influential factors driving the model's results. Figure 9 demonstrates this process using RFR, where the decrease in R^2 helps quantify the predictive value of each feature.

Reaction energy emerges as the most significant feature in the model, exhibiting the highest mean decrease in R^2 , indicating its dominant influence on the model's predictive accuracy. The strong correlation between these two variables highlights the essential role that thermodynamic properties play in determining the energy barrier of a reaction, while activation energy specifically governs the kinetics by dictating the energy barrier that must be overcome for the reaction to proceed. Thus, as it had to be expected, omitting reaction energy from the model would lead to a marked reduction in its predictive capability, underscoring the critical importance of reaction energy in the accurate modeling of activation energy. Reaction energy encapsulates the thermodynamic changes that occur during the chemical transformation of reactants to products. It serves as a fundamental indicator of the catalytic efficiency of a system, reflecting the energy released or consumed in the reaction. The significance of reaction energy in this context suggests a substantial interplay between thermodynamics and activation energy, which is consistent with the Brønsted–Evans–Polanyi principle [91–93]. This principle posits that the activation energy of a reaction is typically related to its reaction energy, that is, more exergonic (thermodynamically favorable) reactions generally exhibit lower activation barriers. This relationship emphasizes that the thermodynamic favorability of a reaction directly influences its kinetic parameters, offering a deeper understanding of the underlying catalytic behavior.

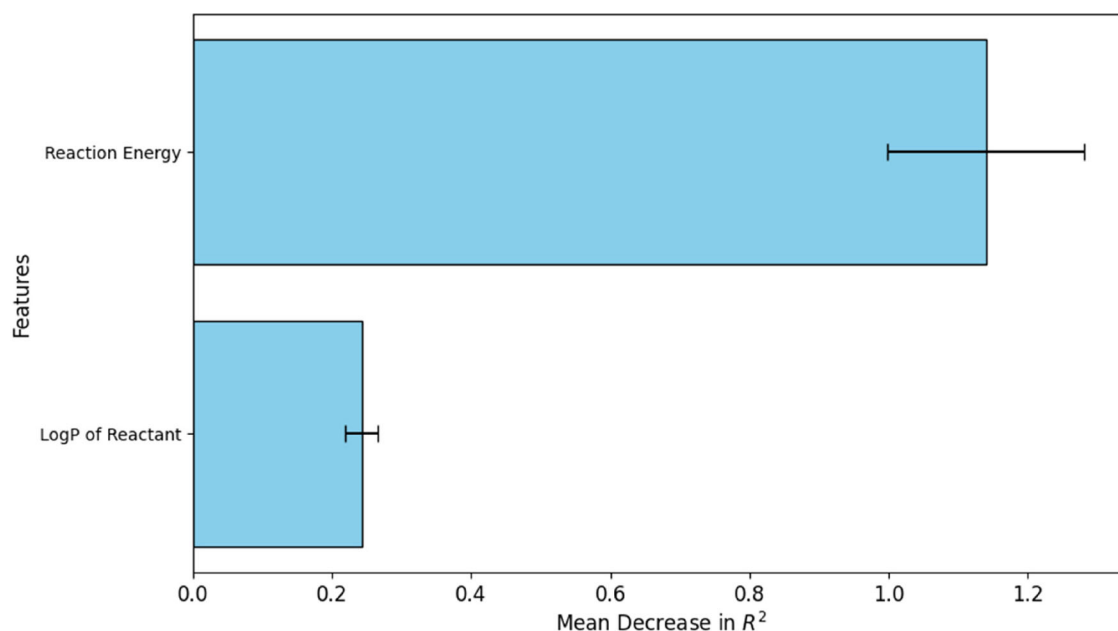


FIGURE 9 | Permutation feature importance analysis using random forest regression.

In addition to feature permutation importance, SHAP analysis was also performed to analyze feature importance and their impact on the output. According to the results presented in Figure 10, the reaction energy is the most important feature, which validates the previous feature permutation importance results. Moreover, according to the average SHAP values, changes in reaction energy move the predicted activation barrier about twice as much, on average, than changes in LogP. According to the SHAP values presented in the graphic below in Figure 10, for the reaction energy (top row), the lower reaction energies (blue) cluster on the left (negative SHAP), meaning that they tend to push the model's predicted barrier down. Whereas, higher reaction energies (red) cluster on the right (positive SHAP), meaning that they tend to result in higher activation energies. For LogP of reactant (bottom row), lower LogP values (blue) tend to give negative SHAP, which means that, in the model, they slightly lower the predicted barrier. Whereas high LogP (red) tends to give positive SHAP (raising the predicted activation energy).

This is very useful for the first-principles calculation of activation energies, since reaction energies are significantly cheaper to calculate computationally. Knowing that a BEP-like relationship holds, reaction energies can then be used to assess (at least preliminarily) the catalytic potential of a MXene surface. While there are exceptions in specific cases [94], this general trend holds true, reinforcing the connection between thermodynamic properties and reaction kinetics. The identification of reaction energy as a key factor in activation energy prediction further establishes it as a crucial bridge between the thermodynamic and kinetic aspects of catalysis, providing valuable insights into the catalytic performance of the studied MXenes.

Fajin et al. [95] identified the co-adsorption energy of OH and H, the products of water dissociation, as the most significant descriptor for a range of surfaces, including monometallic, bimetallic, and trimetallic surfaces, metallic nanotubes, and platinum

nanoparticles. However, this descriptor was found to strongly correlate with the reaction energy of the water dissociation into OH and H (Equation (1)), which was excluded in their study to prevent multicollinearity. Had the reaction energy been used instead of the co-adsorption energy of OH and H, it likely would have emerged as the most significant descriptor, as seen in our current work. Although more computationally demanding, the reaction energy offers a more comprehensive description, accounting for both the initial and final reaction states.

The LogP value of the main adsorbate species (H_2O , OH, H_2 , and CO_2), a measure of the hydrophobicity of the compound, emerges, quite unexpectedly, as the second most significant feature in the model for predicting activation energy, although its influence is secondary to that of reaction energy. This observation suggests that while the hydrophobicity of the reactant, as indicated by its LogP, does impact activation energy, it exerts a comparatively smaller effect than thermodynamic factors like reaction energy. Hydrophobic molecules typically exhibit distinct behaviors in catalytic reactions, particularly in their interactions with catalytic surfaces [96]. The moderate correlation between LogP and activation energy implies that hydrophobicity can influence how reactants interact with the catalyst, thereby affecting the energy barrier of the reaction. Specifically, more hydrophobic reactants may adhere differently to catalyst surfaces compared to more hydrophilic reactants. This difference in adsorption can alter the activation energy, either by stabilizing or destabilizing the transition state, depending on the nature of the interaction. This effect could vary depending on the specific catalytic surface and the environment in which the reaction takes place [97].

Our study primarily aims to predict the activation energies for reactions involving MXenes and the H_2O , OH, CO_2 , and H_2 species (Equations (1)–(4)) because of their importance in the redox route of the WGS reaction. We focused on thermodynamic and hydrophobicity-related factors. Future work incorporating

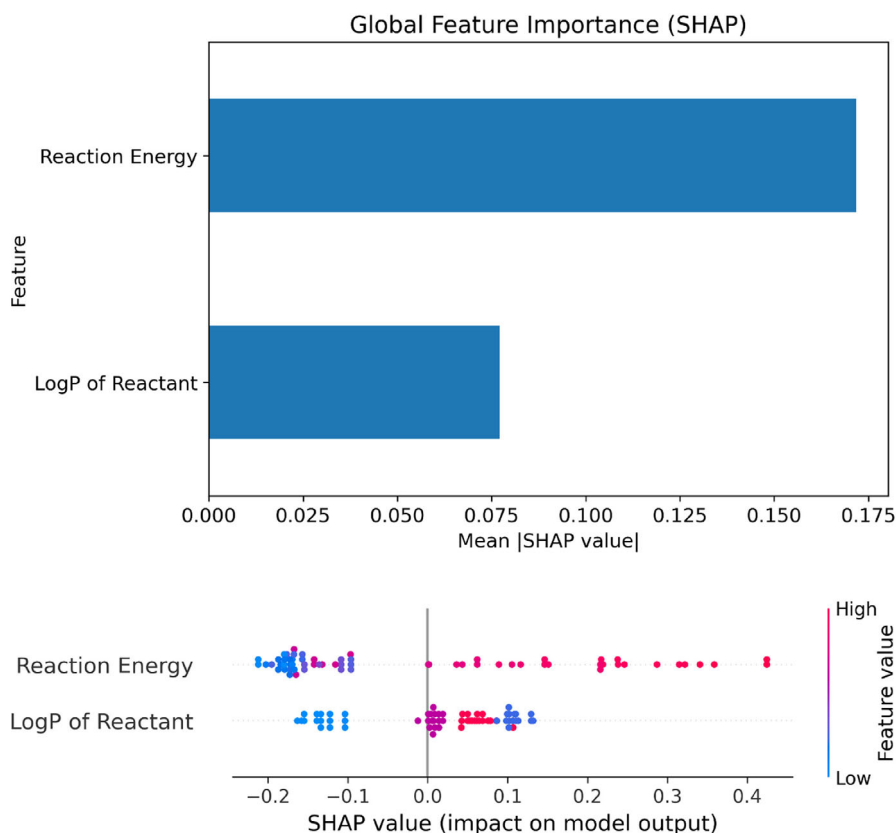


FIGURE 10 | Average SHAP values (above) and SHAP values (below) for the selected descriptors in this work. For the SHAP values, each dot is one sample, plotted at its SHAP value (x-axis) and colored by the actual feature value (blue = lower than average, red = higher than average).

electronic structure features and solvent effects could further enhance the predictive accuracy of our model. Solvent effects modeled using an implicit solvent formalism were found to noticeably affect adsorption energies on metal surfaces and could be relevant, for example, in electrocatalytic reactions [98]. The observed relationship between feature importance and reaction kinetics suggests that a multi-variable approach, integrating thermodynamic, kinetic, and electronic descriptors, may provide a more comprehensive understanding of catalytic behavior. Expanding feature selection to include these aspects could yield deeper insights into the mechanistic foundations of activation energy variations, ultimately aiding in the rational design of high-performance catalysts. Beyond these factors, charge distribution on the MXene surface and the distance between active sites may influence activation energy predictions by affecting reactant adsorption and transition state stabilization [99].

Adsorption energy, a key descriptor in heterogeneous catalysis, could further strengthen predictive models by quantifying reactant–catalyst interaction strength. Additionally, incorporating reactant-specific properties, such as dipole moment, polarizability, and electron affinity, could offer deeper insights into how molecular characteristics shape catalytic performance. In this study, we used a single predictive model for all reactions. However, for future work, it may be useful to apply multitask learning or hierarchical modeling approaches. These methods could help better separate features that are specific to each reaction from those that are shared across all reactions. This would allow the model to better capture both general trends and unique

patterns, especially when working with a larger set of diverse reactions or more complex catalytic systems.

4 | Conclusions and Perspectives

This study underscores the potential of machine learning methods, particularly ensemble techniques like GBR and RFR, in predicting the activation energies of reactions involving MXenes. Among these models, RFR demonstrated the best performance, achieving an impressive mean cross-validation R^2 of 0.95, with a root mean square error of 0.17 eV and a mean absolute error of 0.12 eV. These results confirm the efficacy of RFR in accurately predicting activation energies, highlighting its importance as a predictive tool. Feature selection revealed that two key attributes, namely, the reaction energy and the logarithm of the partition coefficient of the reactant, are the most influential in determining the activation energy. Notably, reaction energy was identified as the most important feature in influencing the model's predictions. This insight is crucial for understanding the key factors that drive activation energies of reactions occurring on MXene surfaces.

Future research would benefit from incorporating additional data points from DFT calculations, expanding the dataset to cover a broader range of elementary reactions relevant to catalytic processes. This would deepen the understanding of MXene catalysis and improve the generalizability and robustness of machine learning models, leading to more accurate predictions

across various chemical reactions. Integrating additional features, such as reactant-to-surface distances, could further refine these models by capturing subtle catalytic effects. In this context, particular attention should be given to hydrogen carrier molecules like methanol (CH_3OH), formaldehyde (H_2CO), formic acid (HCOOH), and ammonia (NH_3), which are essential in energy-related and industrial applications. A diverse selection of reactants will enhance the evaluation of MXenes as catalysts, providing a more comprehensive understanding of their catalytic performance.

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Conflicts of Interest

There are no conflicts of interest to declare.

Data Availability Statement

The data set used in this work, including experimental data and all the calculated features, is available in Mendeley Data as an Excel spreadsheet file. The underlying machine learning code for this study is also available as a Python script in Mendeley Data. Both files can be accessed via this link: <https://doi.org/10.17632/4zmkpw3wxx.2>.

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Supporting Information

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